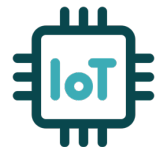
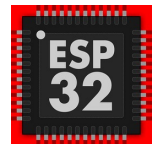


Node-RED:

2. Overwrite Payload with Function Node

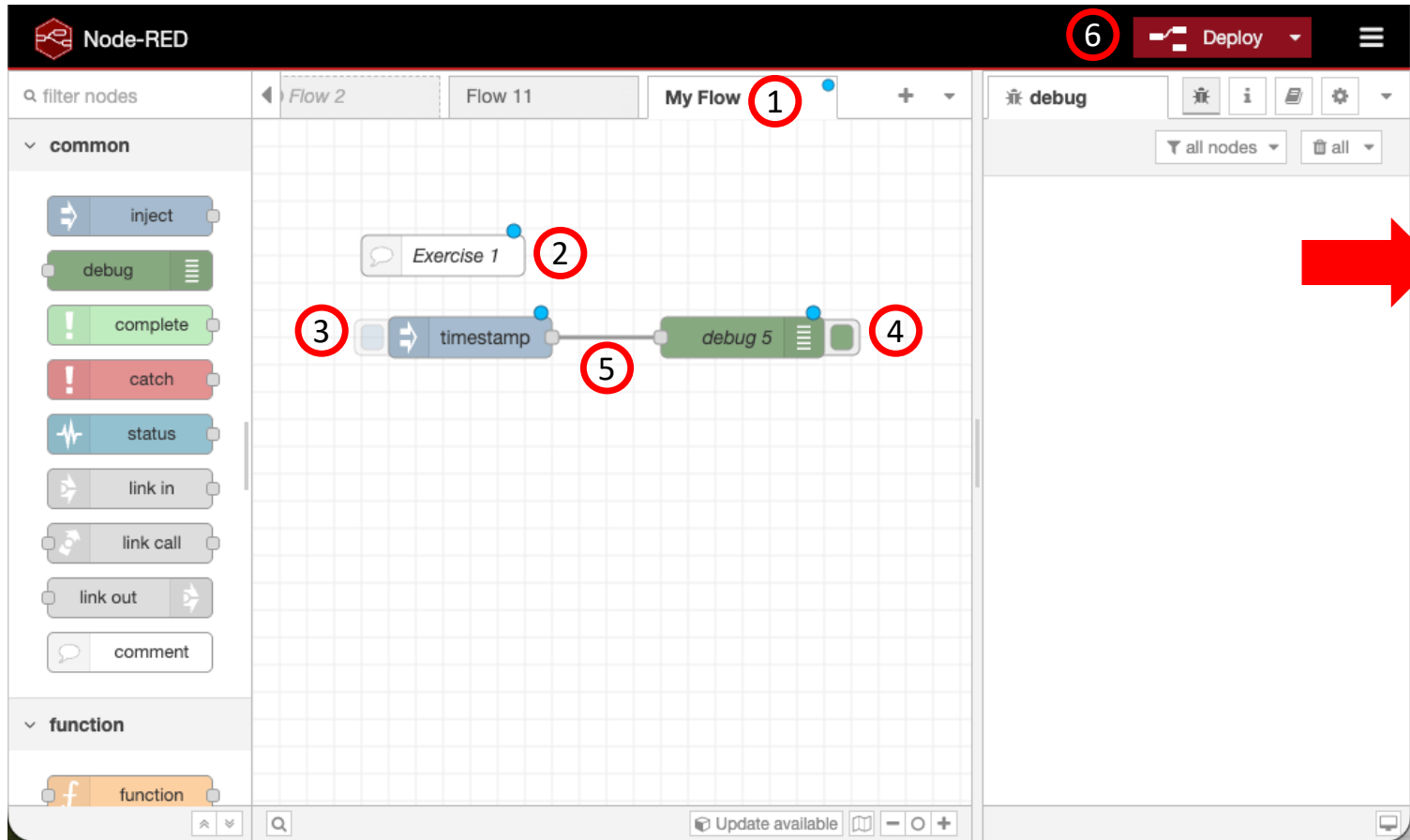


By the end of this section, students should be able to:

- i. 1 inject with 1 debug node
- ii. 1 inject node + 1 function node + 1 debug node
- iii. 3 injects nodes + 1 function node + 1 debug node
- iv. Function with two outputs

UniKL BMI Comm Tech Section
@ TVET MARA - MJII
SEPT 2026

i. 1 inject node with 1 debug node



The screenshot shows the Node-RED interface with a flow named "My Flow" selected. The flow contains three nodes: an inject node (labeled "Exercise 1"), a timestamp node, and a debug node (labeled "debug 5"). The nodes are connected in sequence. The inject node is numbered 1, the timestamp node is numbered 2, and the debug node is numbered 3. The flow is numbered 4, 5, and 6. A red arrow points from the main interface to a zoomed-in view of the debug node output.

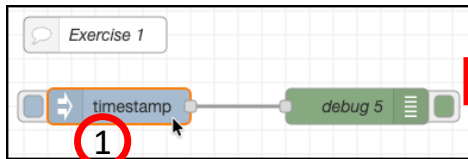
The zoomed-in view of the debug node output shows the following message:

```
9/13/2026, 8:07:22 PM node: debug 5
msg.payload : number
1789301242362
```

Node-RED: 2. Overwrite Payload with Function Node

i. 1 inject node with 1 debug node → Change properties

Double click timestamp node



Edit Inject node

Delete Cancel Done

Properties

Name: Name

msg. payload = milliseconds since epoch

msg. topic = a_z

+ add inject now

☐ Inject once after 0.1 seconds, then

Repeat: none

Enabled

milliseconds since epoch

flow.

global.

a_z string

0_9 number

boolean

{ } JSON

01 10 buffer

timestamp

: expression

\$ env variable

msg.

inject now

Repeat

none

interval

interval between times

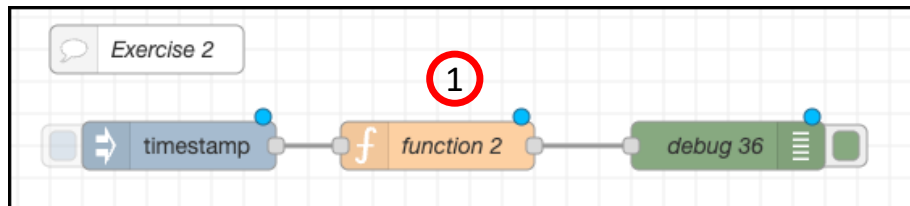
at a specific time

Enabled

Self Challenge.

Change payload to string and put your name next to it.

ii. 1 inject node + 1 function node + 1 debug node



```
Setup On Start On Message On Stop

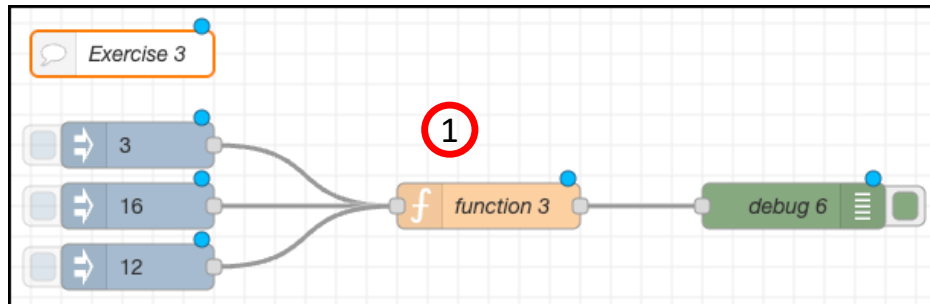
1 msg = {
2   "payload":{
3     "temperature":27,
4     "humidity":70
5   }
6 }
7 return msg;
```

```
1 msg = {
2   "payload":{
3     "temperature":27,
4     "humidity":70,
5     "myMessage": "HELLO TVET MARA-MJII"
6   }
7 }
8 return msg;
```

The debug console shows the output of the function node. It displays the message object with the following structure:

```
msg.payload : Object
  {
    temperature: 27,
    humidity: 70,
    myMessage: "HELLO TVET MARA-MJII"
  }
```

iii. 3 inject nodes + 1 function node + 1 debug node



```
1 if (msg.payload >5){
2     msg.payload="greater than 5" ;
3     return msg;
4 } else {
5     // nothing happens
6 }
7
```

Try this

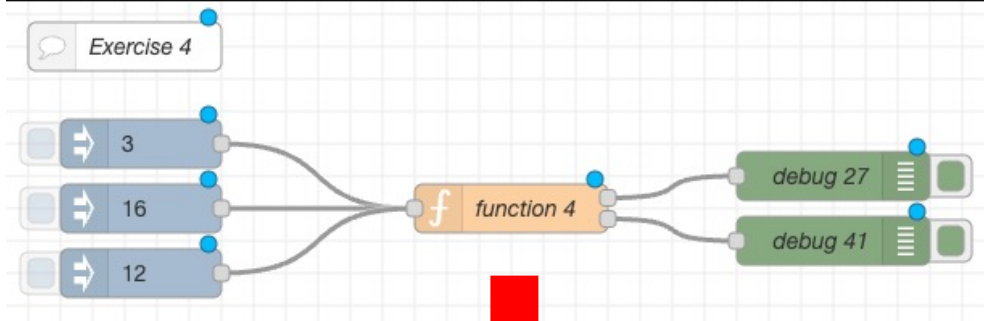
```
var value = msg.payload;
if (value > 5) {
    msg.payload = value + " is greater than 5";
} else {
    msg.payload = value + " is lower than or equal to 5";
}
return msg;
```

Deploy

9/13/2026, 10:50:33 PM node: debug 6
msg.payload : string[14]
"greater than 5"

9/13/2026, 10:50:36 PM node: debug 6
msg.payload : string[14]
"greater than 5"

iv. function with two outputs



```
1 if (msg.payload <5){  
2     return [msg, null];  
3 } else if (msg.payload >9){  
4     return [null, msg];  
5 }
```

Properties

Name: function 4

Setup: On Start, On Message, On Stop

Outputs: 2

Timeout: 0

Modules: Module name, Import as

Deploy

9/13/2026, 11:17:28 PM node: debug 27
msg.payload : number
3

9/13/2026, 11:17:28 PM node: debug 41
msg.payload : number
16

9/13/2026, 11:17:30 PM node: debug 41
msg.payload : number
12

END